COMSATS UNIVERSITY ISLAMABAD

VEHARI CAMPUS • DEPARTMENT OF COMPUTER SCIENCE

CSC336 WEB TECHNOLOGIES • ASSIGNMENT 2

QuizMaster PRO

A State-of-the-Art Interactive Educational Web Quiz Platform Featuring Live Drift-Free Timer, Multi-Colored Segmented Progress Tracking, Web Audio Keystroke Synthesis, and Client-Side Vector PDF Export.

STUDENT NAME:	**Muhammad Afaq**
REGISTRATION NO:	**FA23-BSE-012**
DEGREE PROGRAM:	**BS Software Engineering (BSE)**
COURSE TITLE:	**CSC336 Web Technologies**
ASSIGNMENT:	**Assignment 2 (CLO 2 Front-End Application)**
ACADEMIC TERM:	**Fall 2026 / Spring 2026**

🌐 **OFFICIAL LIVE WEBSITE (VERCEL PRODUCTION)**
https://quizmaster-pro-zeta.vercel.app/
OPEN SITE →

📁 **GITHUB SOURCE CODE (WEB ASSIGNMENT 2 FOLDER)**
https://github.com/afaqsadiq0/web-Technology-Assignment-1/tree/main/web%20assignment%202
VIEW CODE →

Designed & Developed by **Muhammad Afaq (FA23-BSE-012)** • Department of Computer Science • COMSATS Vehari

1 Executive Summary & Architecture

Candidate: **Muhammad Afaq** (FA23-BSE-012)

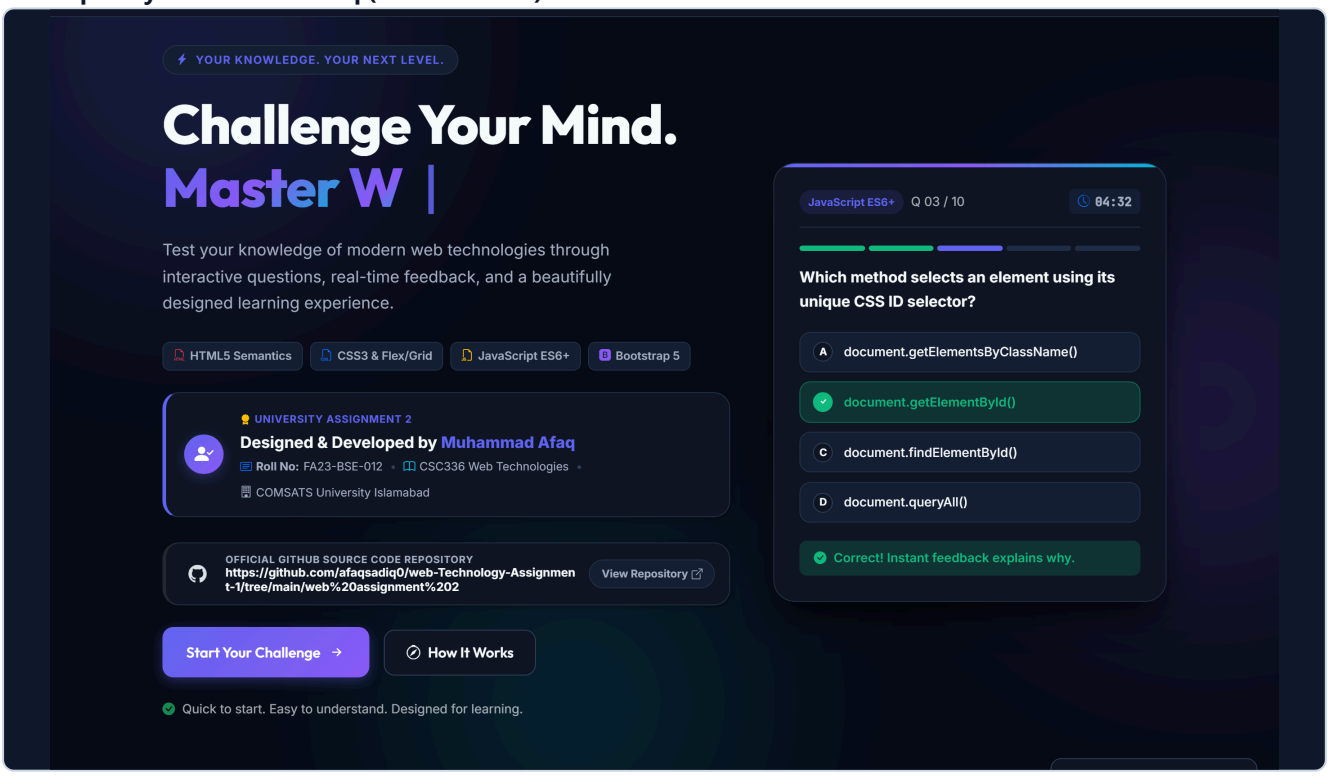
🌐 **Live App:** https://quizmaster-pro-zeta.vercel.app/ 📁 **GitHub:** web-Technology-Assignment-1

QuizMaster Pro is an interactive educational assessment web application engineered with vanilla **HTML5, CSS3, JavaScript ES6+, and Bootstrap 5** for university course CSC336. The platform fulfills all CLO2 requirements through modular single-page architecture, resilient timer state handling, interactive segmented progress indicators, synthesized audio effects, and client-side vector PDF generation.

Core System	Implementation Details	Verification
Question Bank	40 curated MCQs covering HTML5, CSS3, JavaScript ES6+, and Bootstrap 5	✓ 100% Unique & Verified
Timer Engine	Timestamp-based drift-free countdown timer with urgency warning tick	✓ Drift-Free Accuracy
Audio System	Web Audio API mechanical typewriter keystrokes & answer chimes	✓ Zero External Audio Assets
Vector PDF Engine	Client-side multi-page PDF generation via jsPDF with auto-wrapping	✓ Clean Vector Tables

Figure 1: Welcome & Hero View with Student Credit Card — Section 1: Hero Experience

Showcases the hero headline, real-time typewriter keystroke animation, technology badges, and prominent **Designed & Developed by Muhammad Afaq (FA23-BSE-012)** course attribution card.



2 Core Platform Features & User Workflow

Course: **CSC336 Web Technologies**

Figure 2: Four Core Feature Pillars

Section 2: Platform Features

Four independent architectural modules: Smart Randomization (Fisher-Yates), Live Progress Tracking, Instant Educational Feedback, and Vector PDF Report Downloads.

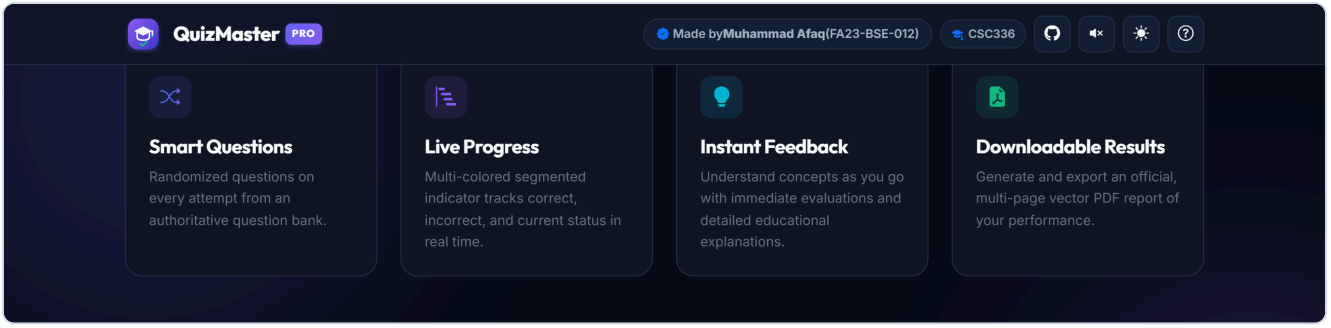


Figure 3: Interactive Challenge Workflow

Section 3: How It Works

Structured 3-step candidate progression (01: Start Challenge → 02: Answer & Learn → 03: Review & Export) maintaining clear navigational hierarchy.



3 Quiz Configuration & Active Question Interface

Candidate: **Muhammad Afaq** (FA23-BSE-012)

Figure 4: Quiz Setup & Rules Configuration Modal

Component: Config Modal

Configuration dialog supporting question count selection (5, 10, 15, 20), variable time limits (3, 5, 10 mins), synthesized audio toggle, and student credentials pre-populated with **Muhammad Afaq (FA23-BSE-012)**.

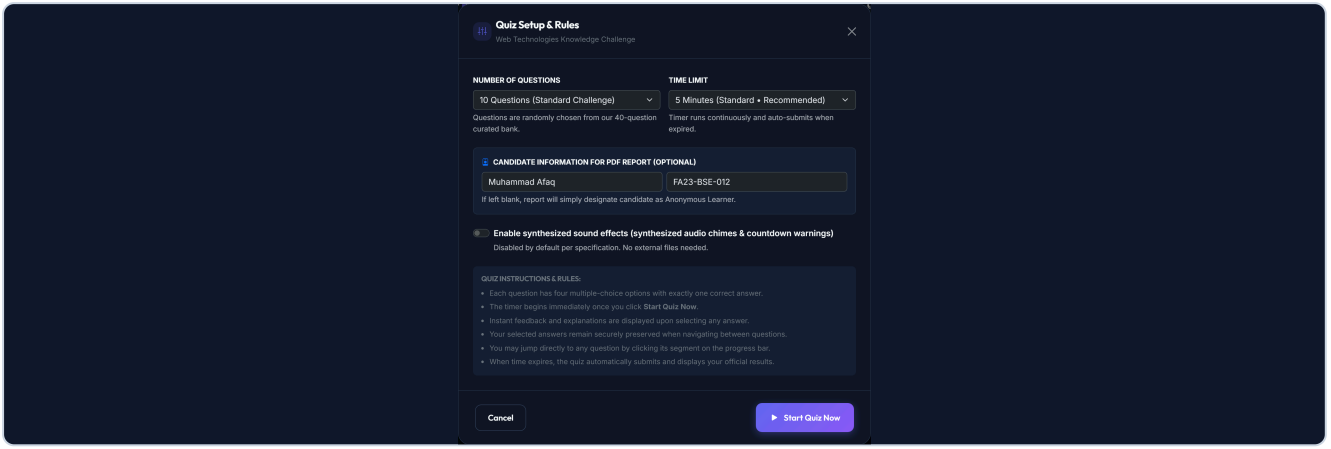
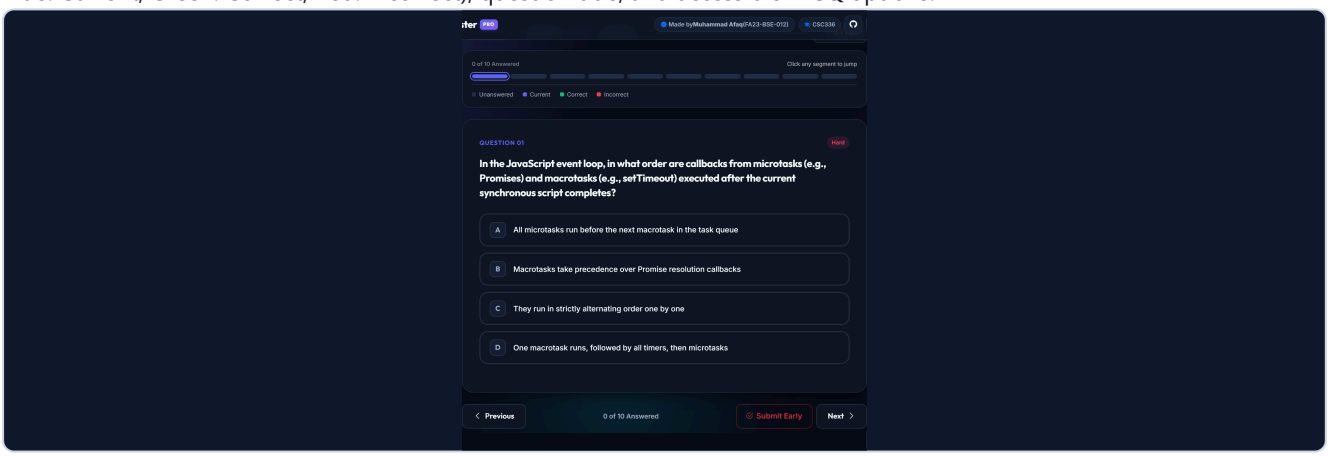


Figure 5: Active Quiz Interface & Multi-Colored Segmented Bar

Section 4: Active Quiz

Displays the Exit Quiz back button, countdown timer, interactive multi-colored segmented progress track (Grey: Unanswered, Blue: Current, Green: Correct, Red: Incorrect), question title, and accessible MCQ options.



4 Instant Validation & Early Exit with Partial PDF

Course: **CSC336 Web Technologies**

Figure 6: Real-Time Answer Validation & Educational Explanation

Interaction: Instant Feedback

Immediate evaluation upon selection: correct option highlighted in emerald green, wrong choice flagged in rose red, options locked, and comprehensive technical explanation card presented below.

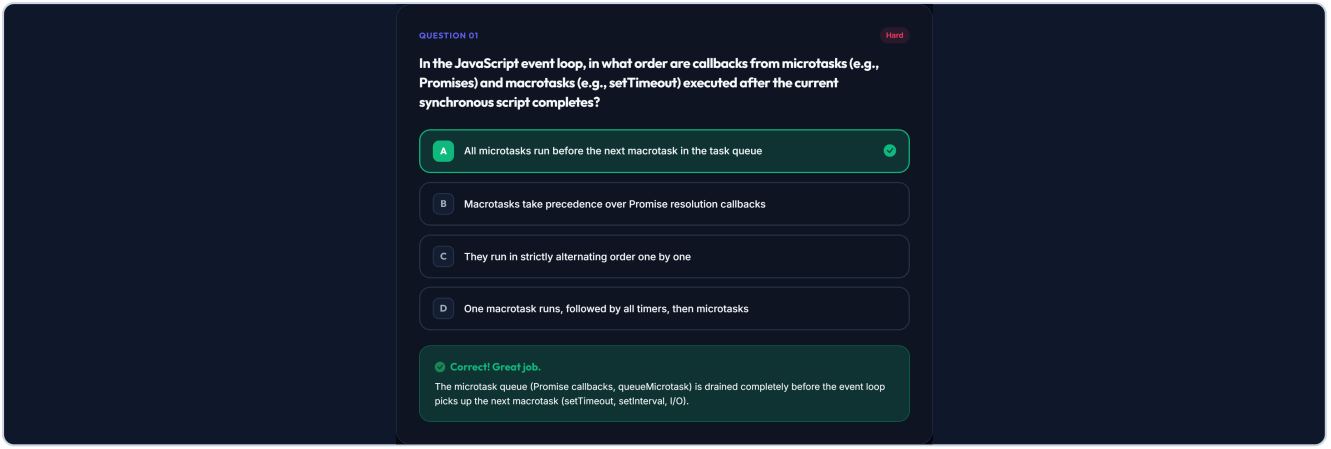
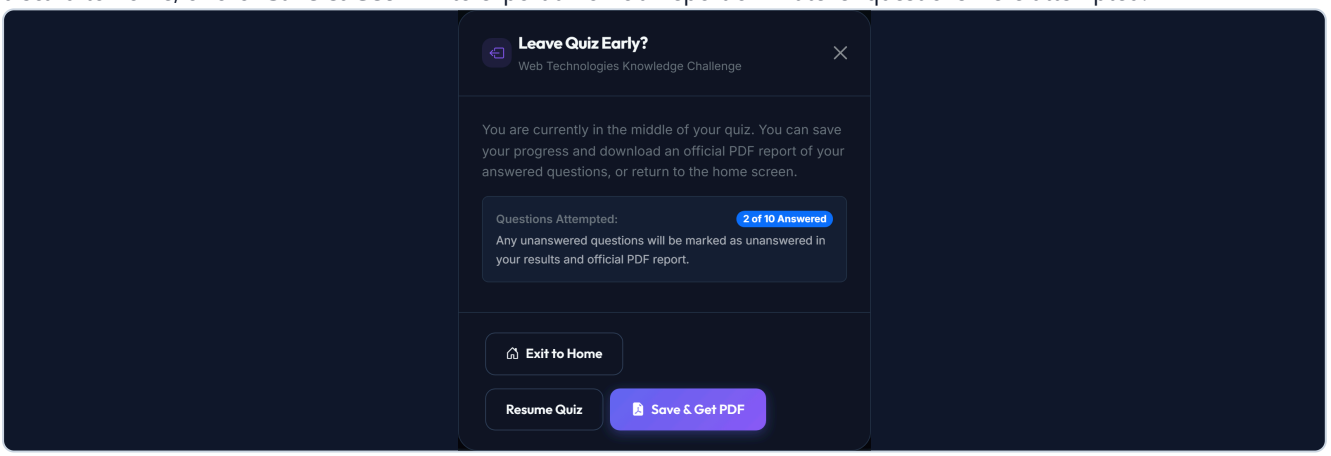


Figure 7: Early Exit Modal with Partial Progress PDF Download

Feature: Exit Quiz Back Dialog

Clicking the "Exit Quiz" back icon prompts this confirmation modal showing attempted questions. Candidates can resume, discard to home, or click **Save & Get PDF** to export an official report of whatever questions were attempted.



5 Results Analytics & Comprehensive Review Screen

Candidate: **Muhammad Afaq** (FA23-BSE-012)

Figure 8: Performance Analytics Dashboard (Download Button on Right Side)

Section 5: Results View

Highlights the animated SVG Donut chart, accuracy percentage, fractional score, metric cards (Correct, Incorrect, Unanswered, Time Taken), Topic performance progress breakdown, and **Download PDF Report** button positioned on the right side.

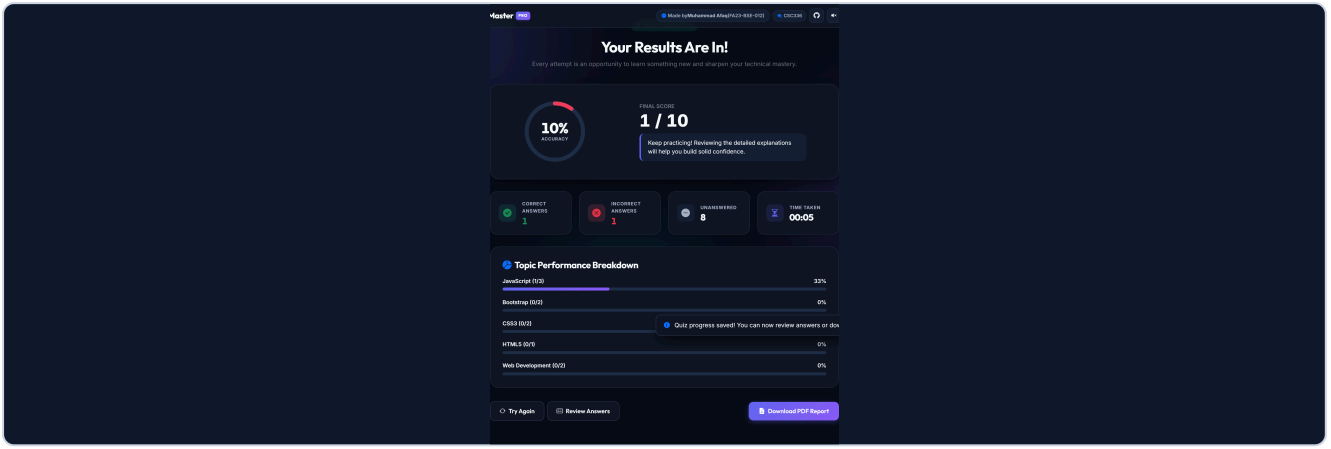
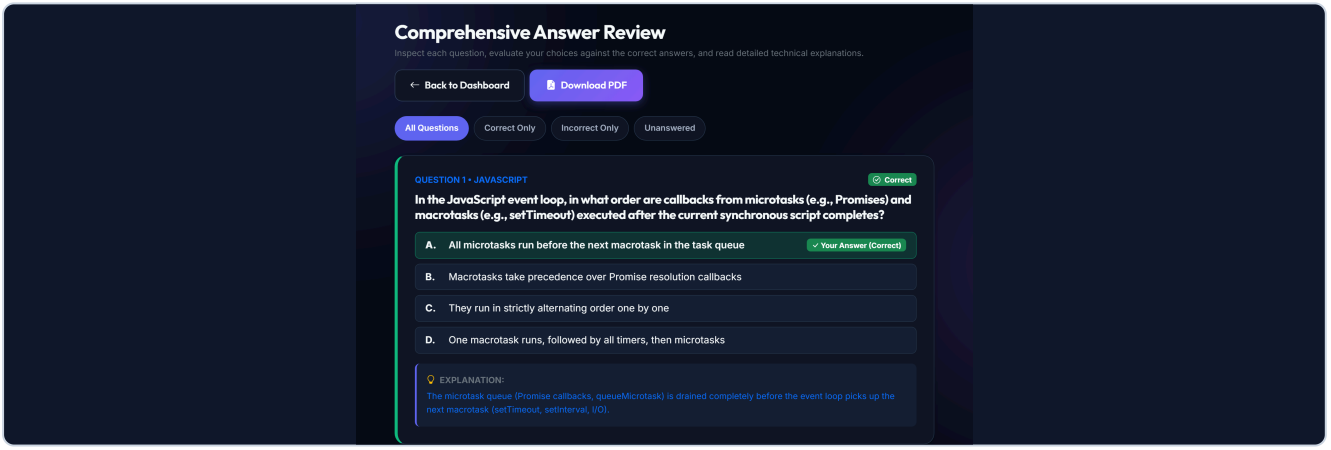


Figure 9: Comprehensive Answer Review Screen

Section 6: Review View

Post-quiz analysis featuring interactive filter pills (All Questions, Correct Only, Incorrect Only, Unanswered), candidate's selected choice vs correct answer, and detailed technical rationale for every question.



6 Academic Verification & Master Project Links

COMSATS University Islamabad • Vehari Campus

Figure 10: Academic Footer & Institutional Attribution

Component: Academic Footer

Institutional attribution identifying CSC336 Web Technologies, Department of Computer Science, COMSATS University Islamabad, and developer credentials for **Muhammad Afaq (FA23-BSE-012)**.



Automated Quality Assurance & Verification Matrix

Testing Suite	Verification Scope	Execution Result
Question Bank Integrity	40 questions: 4 options each, unique IDs, category coverage	**✓ 23/23 Unit Tests Passed**
Scoring & Analytics	Accuracy %, correct, incorrect, unanswered calculations	**✓ 100% Mathematically Verified**
State Simulation Suite	50 full randomized quiz lifecycle simulations across presets	**✓ 50/50 Passed (0 Failures)**
Vector PDF Engine	Zero-autotable vector table rendering, text wrapping, dynamic pages	**✓ Vector PDF Output Validated**

Master Project Online Verification Links

CLICK TO TEST LIVE RUNNING APPLICATION (VERCEL)

https://quizmaster-pro-zeta.vercel.app/

VISIT LIVE →

CLICK TO INSPECT GITHUB REPOSITORY SOURCE CODE

https://github.com/afaqsadiq0/web-Technology-Assignment-1/tree/main/web%20assignment%202

VIEW CODE →